Activity 4.2: Writing Inequalities from Graphs & Situations

A

Name: _____ Partner's name: _____ Date: _____

Directions: Write the inequality described in each problem. Use x and y as the variables.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

A dashed line passes through $(0, 2)$ and $(1, 5)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

2

A solid line has slope -1 and y-intercept 4. The region below the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

3

A solid line passes through $(0, -1)$ and $(2, 0)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

4

A dashed vertical line passes through $(4, 0)$. The region to the left is shaded.

Answer: _____ Match? ☐ Yes ☐ No

5

A solid horizontal line passes through $(0, -3)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

6

The boundary is the dashed line $y = 2x - 3$. The point $(0, 0)$ is in the shaded region.

Answer: _____ Match? ☐ Yes ☐ No

7

You can spend at most \$40 on apples at \$2 per pound (x) and cherries at \$5 per pound (y).

Answer: _____ Match? ☐ Yes ☐ No

8

You want to raise at least \$400 selling adult tickets at \$8 (x) and child tickets at \$5 (y).

Answer: _____ Match? ☐ Yes ☐ No

Activity 4.2: Writing Inequalities from Graphs & Situations

B

Name: _____ Partner's name: _____ Date: _____

Directions: Write the inequality described in each problem. Use x and y as the variables.

Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.

1

A dashed line passes through $(0, 2)$ and $(2, 8)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

2

A solid line passes through $(0, 4)$ and $(4, 0)$. The region below the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

3

A solid line passes through $(-2, -2)$ and $(4, 1)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

4

A dashed vertical line passes through $(4, -3)$. The region to the left is shaded.

Answer: _____ Match? ☐ Yes ☐ No

5

A solid horizontal line passes through $(5, -3)$. The region above the line is shaded.

Answer: _____ Match? ☐ Yes ☐ No

6

The boundary is the dashed line $y = 2x - 3$. The point $(1, 5)$ is in the shaded region.

Answer: _____ Match? ☐ Yes ☐ No

7

You can spend at most \$40 on pens at \$2 each (x) and notebooks at \$5 each (y).

Answer: _____ Match? ☐ Yes ☐ No

8

You want to raise at least \$400 selling small candles at \$8 each (x) and large candles at \$5 each (y).

Answer: _____ Match? ☐ Yes ☐ No